12.8 Calculating Percent Error and Simple Interest

Lesson Introduction

Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.	Learning Targets	• I can find percent error. • I can solve problems that involve simple interest.

	Building On	Working Toward
Benchmark Coherence	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.	N/A

Essential Questions	• What is percent error? • How do you calculate percent error?
Lesson Narrative	After the 12.8.1 Warm-Up, students begin with an Exploration where they compare the actual versus estimated amounts in different contexts using a variety of strategies learned throughout the unit. This leads into introducing them to the concept of percent error, how to calculate it, and how to use it to solve real-world problems. Students then work with their peers and then independently to practice applying this in context. The lesson culminates with students reflecting on unit learning targets.

Component Summary	12.8.1 Warm-Up (~5 mins.)	Determine the better deal based on discounts *(SE p. 254)*	12.8.4 Guided Practice (8-10 mins.)	Practice using percent error to solve real-world problems *(SE p. 256)*
	12.8.2 Exploration (7-10 mins.)	Solve real-world percent increase and decrease problems *(SE p. 254)*	12.8.5 Guided Instruction (10-15 mins.)	Develop an understanding of simple interest *(SE p. 257)*
	12.8.3 Guided Instruction (10 mins.)	Learn how to find percent error *(SE p. 255)*	12.8.6 Wrap-up (~5 mins.)	Reflect on understanding of unit learning targets *(SE p. 259)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Percent error • Simple interest <u>Additional Terms:</u> N/A	N/A	• Consider sharing the focus of today's lesson with grade-level science teachers, as the teachers may be able to help you make connections to how scientists use percent error.

Teacher Tips	Common Misconceptions	Things to Consider
	• Students may struggle to differentiate between the actual value and the intended value when finding the percent error. • Students may confuse the concepts of percent error and percent change. Use this as an opportunity to connect the concepts and highlight their differences. • Students may try to round answers when not prompted to do so.	• Encourage students to highlight and label the important values in the problem. When doing so, be sure that students are still focusing on the context and the meaning of the values in it by prompting them to discuss the contexts and the meaning of percent error in them. • Allowing students to use calculators during the lesson will help students focus on discovery and on the process of calculating percent of change. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers. • Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common, careless errors that many students make.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share In 12.8.4 Guided Practice, have students initially solve the problems independently. Then have students discuss their work. These discussions will help students gain confidence and understanding.	**MA.K12.MTR.3.1: Mathematical Fluency** In 12.8.5 Your Turn!, students practice solving percent error problems. As students practice, they will develop efficient and generalizable mathematical methods based on what they've been learning throughout the unit.
Differentiate Instruction	Encourage students to use the Mark-Up strategy to make annotations as they read problems. Ensure these annotations go beyond simply circling numbers to include markings related to the meaning of the values in the context.	

Lesson Notes:

12.8.1 Warm-Up: Would You Rather?

Component Overview: *Students compare two options for ordering pizza and decide which they would rather take.*

12.8.1 Warm-Up: Would You Rather?

It's "Multiple Monday" and the more pizzas purchased, the greater the discount applied. If one pizza is ordered, the discount is 10%. If two pizzas are ordered, the discount is 20%, and for three pizzas, the discount goes up to 30%.

1. Would you rather order 3 pizzas on separate checks, or order 3 pizzas on the same check and split the cost?
   I would rather have 3 pizzas on the same check.

2. Justify your answer using mathematics.
   If the pizzas were $10, they would be $9 each if ordered separately. If ordered together, each pizza would be $7.

Ask students if they have seen discounts at their local pizzeria. Bring in sales flyers from pizzerias to show students the applications of the math they are learning.

12.8.2 Exploration: Estimate vs. Actual

Component Overview: *Students explore situations where the recommended or estimated value is different from the actual value. This component sets students up to be introduced to percent error in the next component.*

12.8.2 Exploration: Estimate vs. Actual

1. Instructions to care for a plant say, "Water with $\frac{3}{4}$ cup of water daily." The plant has been getting 25% too much water. How much water has the plant been getting? Include units.
   $\frac{3}{4} \times \frac{5}{4} = \frac{15}{16}$ *cups*

2. The pressure on a bicycle tire is 63 psi. This is 5% higher than what the manual states is the correct pressure. What is the correct pressure? Include units.
   $1.05x = 63;\ x = 60$ *psi*

3. The crowd at a sporting event is estimated to be $2{,}500$ attendees. The exact attendance is $2{,}486$ people. By what percentage is the estimated attendance off?
   $\frac{2{,}500 - 2{,}486}{2486} = \frac{14}{2486} = -0.0056 = -0.56\%$

4. Discuss with your partner what strategies you used to solve questions $1 - 3$.
   Answers will vary. For the first problem, I multiplied the recommended amount by a fraction equivalent to 125% to determine the amount of water the plant has been getting. For the second problem, I set up a percent equation and solved it to find the correct original pressure. For the third problem, I used the percent of change formula.

As students work through the problems, listen to students' discussions and elicit responses about the processes they are using to solve problems.

Monitor that students are not rounding inappropriately, as none of the questions say to round.

Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common, careless errors that many students make.

There are multiple strategies for solving the problems in this component. Encourage students to use strategies that make sense for them as they build fluency in solving problems.

12.8.3 Guided Instruction: Percent Error

Component Overview: *Students learn what percent error is and how to calculate it. Students use the percent error formula to solve several problems. Students discuss why the percent error is expressed as a positive percentage.*

12.8.3 Guided Instruction: Percent Error

Percent error is the difference between the estimated number and the actual number as a percentage of the actual value.

Percent error can be used to describe any situation where there is a correct value and an incorrect value, and to describe the relative difference between them.

1. A teacher estimates there are approximately 600 students at his school, but in actuality there are 625 students enrolled.
   a. What is the difference between the teacher's estimate and the actual number of students enrolled at the school?
   25
   b. Using the definition of percent error, finish filling out the ratio of the situation.

$$\left|\frac{\text{Expected Value} - \text{Actual Value}}{\text{Actual Value}}\right| = \left|\frac{-25}{625}\right| = 0.\underline{\ 04\ }$$

   c. The percent error in the teacher's estimate is 4 ____%.
   d. Discuss with your partner why the percent error is positive. Summarize your discussion.
   Percent error is positive because the absolute value is taken after finding the difference of the two values.

2. The front of a cereal box lists the net weight of the product is 21 ounces (oz). After further measurements, it is determined there are actually 23 oz of cereal in the box.
   a. What is the difference, in ounces, between the weight listed on the box and the measured weight?
   2 *ounces*
   b. Find the percent error. Show your work.

$$\left|\frac{21-23}{23}\right| \cdot 100 \approx 8.7\%$$

Consider reading the questions aloud. Encourage students to highlight the important values in the questions and label them as the "expected value" or "actual value." This will make it easier for students to use the formula. Encourage students to explain what the percent error means in the context to ensure they are not simply circling values, but also grasping the context presented.

Prompt students' thinking with the following:

- *What does "net weight" mean? Do you think every box of cereal has the exact weight printed on the package?*
- *Which amount is the expected value?*
- *Which amount is the actual value?*
- *What does the percent error mean in this context?*
- *How does the percent error relate to the percent of change?*

12.8.4 Guided Practice: Percent Error

Component Overview: *Students use percent error to solve real-world problems. Students may solve the problems independently or work with a partner.*

12.8.4 Guided Practice: Percent Error

1. A store owner estimated she would have 125 customers in her store on its opening day. She underestimated the number of customers by 8%. How many customers visited the store on opening day?
   $\frac{x}{125} = \frac{8}{100}$; $x = 10$
   $125 + 10 = 135$ *customers visited the store*

2. To be labeled a jumbo egg, the egg is supposed to weigh 2.5 ounces (oz). Alana buys a carton of jumbo eggs and measures one of the eggs as 2.4 oz. What is the percent error?
   $\frac{0.1}{2.4} \approx 0.0417 = 4.17\%$

3. A student estimates there to be 25.7 milliliters (mL) of a solution in his beaker in science class. He underestimated the amount by 12.5%. What was the actual amount of solution in the beaker? Round to the nearest thousandth.
   $0.875x = 25.7$
   $x = 29.371$ *mL*

4. Lia wanted to take advantage of the pizza special on "Multiple Monday." She estimates it will cost her $\$28$ for two large cheese pizzas after the discount is applied. The actual cost is $\$28.80$. What is the percent error? Round to the nearest whole percent.
   $\frac{0.80}{28.80} \approx 0.0278 \approx 3\%$

Think-Pair-Share
Have students initially solve the problems independently. Then, have students discuss their work. These discussions will help students gain confidence and understanding.

Consider allowing students to select two of the questions to complete. Students will be able to thoughtfully and thoroughly solve the problems without feeling overwhelmed by their workload. They also will be more motivated to solve the problems they select because they will be connected to their interests.

12.8.5 Guided Instruction: Simple Interest

Component Overview: *Students learn what simple interest is, how to calculate it, and how to use it to solve problems in context.*

12.8.5 Guided Instruction: Simple Interest

1. Kristina received $\$100$ in a birthday card and wants to put the money into her savings account, where she will earn an additional 2% each year it remains in her account. This is known as ***simple interest***.

   Simple interest is a method of computing interest. Interest is computed from the (original) principal alone, no matter how much money has accrued so far.

   a. How much is 2% of $\$100$?
   $\$2.00$

   b. How much money will be in Kristina's savings account after one year if she makes no additional deposits or withdrawals after putting in her birthday money?
   $\$102.00$

   c. Compare your answer with your partner and discuss how you determined the answer for Part B.
   I knew that Kristina had $\$100$ already and then earned an additional $\$2$ which equals $\$102$ when added together.

2. Suppose an investment of $\$100$ earns $\$3.50$ interest in one year.

   a. What investment amount will earn $\$61.60$ in two years at the same rate?
   $0.035x = \frac{61.60}{2}$; $x = \$880.00$
   Students could also set up a percent proportion.

Ask students what they know about interest. List examples on the board. Explain that financial institutions charge interest for allowing a customer to borrow money with a loan and pays a customer interest for keeping their money in a savings account. Answer any questions students have about the concept of interest.

b. Ask two other partner pairs to describe how they found the initial investment. Record their initial investment and write your summary of their process. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Initial Investment	My Summary of Their Reason	Initials
$880.00	*Answers will vary. They divided* $61.60 *by* 2 *since there was* 2 *years and then found what amount* $30.80 *is* 3.5% *of.*	
$880.00	*Answers will vary. They found what amount 3.5% is of* $61.60 *and then divided by 2.*	

c. Review the initial investments and reasons provided by other partner pairs. If necessary, fix any mistakes you and your partner made.

Simple Interest Formula

$$I = p \times r \times t$$

where I is the interest earned, p is the principal starting amount of money, r is the interest rate per year, and t is the length of the investment.

3. Tonio took out a six-year car loan for $15,600, with a 3% simple interest rate over the life of the loan.

   a. How much interest, in dollars, will Tonio pay annually?
   $I = (15{,}600)(0.03)(1) = \$468.00$ *per year*

   b. How much will Tonio pay in interest over the course of **six** years?
   $I = (15{,}600)(0.03)(6) = \$2{,}808.00$

Consider asking students the following:

- *How did you determine the initial investment amount for the previous question?*
- *How does the method you used relate to the simple interest formula?*
- *How could you use the formula to determine an interest rate or the length of the investment?*

Consider allowing students to "shop" for a car locally and use the advertised price and a current, realistic percentage rate to see how much interest they would have to pay over the life of a 5-year loan.

12.8.5 Guided Instruction: Simple Interest (continued)

c. What is the total amount Tonio will pay back at the end of his six-year loan?
$15{,}600 + 2{,}808 = \$18{,}408.00$

d. Explain why the total amount Tonio will have paid back by the end of his loan is more than the loan amount he borrowed.
He will pay back more because he has to pay back the amount he borrowed, plus the interest accrued on the loan.

12.8.6 Wrap-Up: Reflection

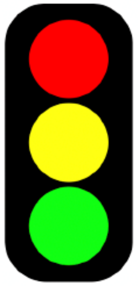
Component Overview: *Students reflect on their understanding of the learning targets from the unit. Encourage students to provide specific details if time allows. Use observations from this reflection to determine where students need support before the summative unit assessment.*

12.8.6 Wrap-Up: Reflection

In this unit, you have learned multiple concepts about percentages and proportional relationships, including those listed below.

- Solving real-world problems using proportions
- Solving discount, markup, tax, tip, and fees problems
- Finding percentage of increase and decrease
- Finding percent error
- Solving problems involving simple interest

1. Use the stoplight reflection below to share how you are feeling about each of the concepts at this point.



I still feel stuck on how to …
Answers will vary. There is no right or wrong answer, this is a student opinion.

I'm starting to get the hang of …
Answers will vary. There is no right or wrong answer, this is a student opinion.

I feel confident and ready to go with …
Answers will vary. There is no right or wrong answer, this is a student opinion.

Some students may struggle to express themselves in writing. Consider providing alternatives, such as using speech-to-text software or an audio recorder (i.e. Flip Grid), or by labeling examples.